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Modifying patterned features using a directional etch

Abstract

Disclosed herein are approaches for modifying patterned features using a directional etch. In one approach, a method may include providing a stack of layers of a semiconductor device, forming an opening through the stack of layers, the opening defined by a first sidewall and a second sidewall, and delivering ions into the first sidewall in a reactive ion etching process. The ions maybe delivered at a first non-zero angle relative to a perpendicular extending from the substrate, wherein the reactive ion etching process removes a first portion of the stack of layers from just a lower section of the first sidewall.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure relates to semiconductor structures and, more particularly, to modifying patterned features using a directional etch.

BACKGROUND OF THE DISCLOSURE

(2) As semiconductor devices continue to scale to smaller dimensions, the ability to pattern features becomes increasingly difficult. These difficulties include in one aspect the ability to obtain features at a target size for a given technology generation. Another difficulty is the ability to obtain varied sidewall shapes of a patterned feature, such as an opening or trench. For example, given a fixed ion beam angle, there is a critical aspect ratio above which the ion beam is blocked from reaching the bottom of the trench. At this point lateral etch sharply drops.

(3) With respect to these and other considerations the present improvements may be useful.

SUMMARY

(4) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended as an aid in determining the scope of the claimed subject matter.

(5) In one aspect, a method may include providing a stack of layers of a semiconductor device, and

forming an opening through the stack of layers, the opening defined by a first sidewall and a second sidewall. The method may further include delivering ions into the first sidewall in a reactive ion etching process, wherein the ions are delivered at a first non-zero angle relative to a perpendicular extending from the substrate, and wherein the reactive ion etching process removes a first portion of the stack of layers from just a lower section of the first sidewall.

(6) In another aspect, a method for patterning opening sidewalls may include providing a stack of layers of a semiconductor device, the stack of layers including a first layer formed over a base layer, a second layer formed over the first layer, and a third layer formed over the second layer. The method may further include forming an opening through the stack of layers, and directing ions into a sidewall of the first layer in a first reactive ion etch process, wherein the ions are delivered through the opening at a first non-zero angle relative to a perpendicular extending from the substrate, wherein the reactive ion etch process removes a lower portion of the sidewall of the first layer, and wherein a plane defined by the sidewall of the first layer is oriented at the first non-zero angle after the lower portion of the sidewall of the first layer is removed.

(7) In yet another aspect, a method may include providing a stack of layers of a semiconductor device, the stack of layers including a first layer formed over a base layer, a second layer formed over the first layer, and a third layer formed over the second layer. The method may further include forming an opening through the stack of layers, and directing ions into a sidewall of the first layer in a first reactive ion etch process, wherein the ions are delivered through the opening at a first non-zero angle relative to a perpendicular extending from the substrate, wherein the reactive ion etch process removes a lower portion of the sidewall of the first layer without removing an upper portion of the sidewall of the first layer, and wherein a plane defined by the sidewall of the first layer is oriented at the first non-zero angle after the lower portion of the sidewall of the first layer is removed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings illustrate exemplary approaches of the disclosure, including the practical application of the principles thereof, as follows:

(2) FIG. **1**A is a top view of a semiconductor device, according to embodiments of the present disclosure;

(3) FIG. **1**B is a cross-sectional side view of the device along cutline X-X′ of FIG. **1**A, according to embodiments of the present disclosure;

(4) FIG. **1**C is a cross-sectional side view of the device of FIG. **1**A along cutline X-X′ following an angled etch, according to embodiments of the present disclosure;

(5) FIG. **2**A is a cross-sectional side view of a device during an angled etch, according to embodiments of the present disclosure;

(6) FIG. **2**B is a cross-sectional side view of the device of FIG. **2**A following the angled etch, according to embodiments of the present disclosure;

(7) FIG. **3**A is a cross-sectional side view of a device during a first angled etch, according to embodiments of the present disclosure;

(8) FIG. **3**B is a cross-sectional side view of the device of FIG. **3**A during a second angled etch, according to embodiments of the present disclosure;

(9) FIG. **3**C is a cross-sectional side view of the device of FIG. **3**B following the angled etch, according to embodiments of the present disclosure;

(10) FIG. **3**D is a cross-sectional side view of the device during a third angled etch, according to embodiments of the present disclosure;

(11) FIG. **3**E is a cross-sectional side view of the device following the third angled etch, according

to embodiments of the present disclosure;
(12) FIG. **4**A is a top view of a device, according to embodiments of the present disclosure;
(13) FIG. **4**B is a cross-sectional side view of the device of FIG. **4**A during a first angled etch, according to embodiments of the present disclosure;
(14) FIG. **4**C is a cross-sectional side view of the device of during a second angled etch, according to embodiments of the present disclosure;
(15) FIG. **4**D is a cross-sectional side view of the device following the second angled etch, according to embodiments of the present disclosure;
(16) FIG. **5**A is a cross-sectional side view of a device during a first angled etch to recess trenches, according to embodiments of the present disclosure;
(17) FIG. **5**B is a cross-sectional side view of the device during a second angled etch to recess the trenches, according to embodiments of the present disclosure;
(18) FIG. **5**C is a cross-sectional side view of the device following the second angled etch to recess the trenches, according to embodiments of the present disclosure;
(19) FIG. **6**A illustrates an exemplary processing apparatus according to embodiments of the disclosure; and
(20) FIG. **6**B depicts details of an exemplary extraction plate according to embodiments of the disclosure.
(21) The drawings are not necessarily to scale. The drawings are merely representations, not intended to portray specific parameters of the disclosure. The drawings are intended to depict exemplary embodiments of the disclosure, and therefore are not to be considered as limiting in scope. In the drawings, like numbering represents like elements.
(22) Furthermore, certain elements in some of the figures may be omitted, or illustrated not-to-scale, for illustrative clarity. The cross-sectional views may be in the form of “slices”, or “near-sighted” cross-sectional views, omitting certain background lines otherwise visible in a “true” cross-sectional view, for illustrative clarity. Furthermore, for clarity, some reference numbers may be omitted in certain drawings.
DETAILED DESCRIPTION
(23) Methods and devices in accordance with the present disclosure will now be described more fully hereinafter with reference to the accompanying drawings, where various embodiments are shown. The methods and devices may be embodied in many different forms and are not to be construed as being limited to the embodiments set forth herein. Instead, these embodiments are provided so the disclosure will be thorough and complete, and will fully convey the scope of the methods to those skilled in the art.
(24) To address the deficiencies of the prior art described above, embodiments of the present disclosure advantageously provide an angled etch to form cavities (e.g., trenches or openings) having non-vertical sidewalls. In some embodiments, multiple etch processes are used to form sidewalls having more than one surface angle. In yet other embodiments, different material layers having different etch rates may be etched at an angle to form sidewalls having more than one surface angle. As a result, the cavities may increase volume and surface area for a conductive (either electrical or heat) fill and/or to increase adhesion area. Furthermore, the angled etch allows manipulation or correction of the sidewall shape, as well as etch depth control without the need for an etch-stop landing layer.
(25) FIGS. **1**A-**1**B depict a portion of a semiconductor device (hereinafter “device”) **100** including a plurality of layers **101** atop a base layer or substrate **102**. As shown, the plurality of layers **101** may include a first layer **104** formed atop the substrate **102**, and a second layer **106** formed atop the first layer **104**. An opening **108** may be formed through the plurality of layers **101**, exposing an upper surface **112** of the substrate **102**. The first layer **104** may be a first material type, while the second layer **106** may be a second material type, different than the first material type. In various embodiments, the materials of the first and second layers **104**, **106** may be any combination of the

following non-limiting materials: SiO.sub.2, a-Si (amorphous Si), spin-on carbon, photoresist, metal photoresist, TiN, doped carbon, or SiN.

(26) As further shown, reactive ions of a first angled etch **114** may be delivered into a first sidewall **116** of the opening **108** at a first non-zero angle ‘ϴ’ relative to a perpendicular **120** extending from a top surface **122** of the second layer **106**. The reactive ions of the first angled etch **114** may also impact a portion of the upper surface **112** of the substrate **102** within the opening **108**, but to minimal effect. Although non-limiting, the reactive ions of the first angled etch **114** may include F, Cl, Br, and/or others.

(27) As shown in FIG. **1**C, the first angled etch **114** removes a first portion **123** (FIG. **1**B) of the stack of layers **101** from just a lower portion **124** of the first layer **104** of the first sidewall **116**. Because the first and second layers **104**, **106** each have a different etch selectivity, the first layer **104** is etched but the second layer **106** generally is not. An upper section **128** of the first sidewall **116** is generally less affected by the first angled etch **114** due to shadowing from the second layer **106**. As a result, a plane defined by the sidewall **116** of the first layer **104** is oriented at the first non-zero angle ‘ϴ’ after the lower portion **124** of the first layer **104** is removed. In other embodiments, the plane defined by the sidewall **116** of the first layer **104** may be oriented at a second non-zero angle relative to the perpendicular **120**. A width (e.g., along the x-direction) of the opening **108** is greater adjacent the upper surface **112** of the substrate **102** than adjacent the second layer **106**. Said another way, a notch or indentation is therefore present in the first layer **104**, tucked beneath the second layer **106**.

(28) FIG. **2**A depicts a portion of another semiconductor device (hereinafter “device”) **300** according to embodiments of the present disclosure. The device **300** may include a plurality of layers **301** atop a substrate **302**. As shown, the plurality of layers **301** may include a first layer **304** formed atop the substrate **302**, a second layer **306** formed atop the first layer **304**, a third layer **307** formed atop the second layer **306**, and a fourth layer **309** formed atop the third layer **307**. An opening **308** may be formed through the plurality of layers **301**. The first layer **304** may be a first material type, the second layer **306** may be a second material type, the third layer **307** may be a third material type, and the fourth layer **309** may be a fourth material type. In this embodiment, the first layer **304** and the third layer **307** are different materials. As such, the first layer **304** and the third layer **307** may have different etch rates. In various embodiments, the materials of the plurality of layers **301** may be any combination of the following non-limiting materials: SiO.sub.2, a-Si (amorphous Si), spin-on carbon, photoresist, metal photoresist, TiN, doped carbon, or SiN.

(29) As further shown, reactive ions of a first angled etch **314** may be delivered into a first sidewall **316** of the opening **308** at a first non-zero angle ‘ϴ’ relative to a perpendicular **320** extending from a top surface **322** of the fourth layer **309**. In the embodiment shown, the reactive ions of the first angled etch **314** may be delivered to each of the exposed plurality of layers **301**. Although non-limiting, the reactive ions of the first angled etch **314** may include F, Cl, Br, and/or others.

(30) As shown in FIG. **2**B, the first reactive ion etch **314** removes a portion of material from just a lower portion **324** of the first layer **304** and the third layer **307**. An upper section **328** of the first and third layers **304**, **307** is generally less affected by the first reactive ion etch **314** due to shadowing from the second layer **306** and the fourth layer **309**, respectively. Because the first and third layers **304**, **307** each have a different etch selectivity relative to the second layer **306** and the fourth layer **309**, the first layer **304** and the third layer **307** are etched while the second and fourth layers **306**, **309** generally are not. However, the first and third layers **304**, **307** are not etched equally. As a result, after the lower portion **324** of the first and second layers **304**, **307** is removed, a first plane defined by the sidewall **316** along the first layer **304** and a second plane defined by the sidewall **316** along the third layer **307** are oriented at different non-zero angles relative to the perpendicular **320**.

(31) Although not shown, a second reactive ion etch may be performed along a second sidewall **317** of the opening **308**, similar to the first reactive ion etch **314**. That is, ions of the second reactive

ion etch may be delivered into the second sidewall **317** at a second non-zero angle relative to the perpendicular **320**. The second reactive ion etching process removes a portion of material from a lower section of the first and third layers **304**, **307** without removing material from an upper section of the first and third layers **304**, **307**. In some embodiments, following the second reactive ion etch, the first sidewall **316** mirrors the second sidewall **317**.

(32) FIG. **3**A depicts a portion of another semiconductor device (hereinafter "device") **400** according to embodiments of the present disclosure. The device **400** may include a plurality of layers **401** atop a substrate **402**. As shown, the plurality of layers **401** may include a first layer **404** formed atop the substrate **402**, a second layer **406** formed atop the first layer **404**, a third layer **407** formed atop the second layer **406**, and a fourth layer **409** formed atop the third layer **407**. An opening **408** may be formed through the plurality of layers **401**. In some embodiments, the first layer **404** and the third layer **407** are the same material, while the second layer **406** and the fourth layer **409** are the same material. As such, the first layer **404** and the third layer **407** may have a same etch selectivity, and the second layer **406** and the fourth layer **409** may have a same etch selectivity, different than the etch selectively of the first layer **404** and the third layer **407**.

(33) As further shown, reactive ions of a first angled etch **414** may be delivered into a first sidewall **416** of the opening **408** at a first non-zero angle '⊖' relative to a perpendicular **420** extending from a top surface **422** of the fourth layer **409**. In the embodiment shown, the reactive ions of the first angled etch **414** may be delivered to each of the exposed plurality of layers **401**.

(34) As shown in FIG. **3**B, the first reactive ion etch **414** removes a portion of material from just a lower portion **424** of the first layer **404** and the third layer **407**. An upper section **428** of the first and third layers **404**, **407** is generally less affected by the first reactive ion etch **414** due to shadowing from the second layer **406** and the fourth layer **409**, respectively. Because the first and third layers **404**, **407** have a different etch selectivity relative to the second layer **406** and the fourth layer **409**, the first layer **404** and the third layer **407** are etched while the second and fourth layers **406**, **409** generally are not. As a result, after the lower portion **424** of the first and second layers **404**, **407** are removed, a first plane defined by the sidewall **416** along the first layer **404** and a second plane defined by the sidewall **416** along the third layer **407** are oriented at a non-zero angle (e.g., the first non-zero angle '⊖') relative to the perpendicular **420**. In this embodiment, the first and second planes are oriented at approximately the same angle relative to the perpendicular **420** following the first reactive ion etch **414**. In other embodiments, the first and second planes may be oriented at different angles relative to the perpendicular **420** following the first reactive ion etch **414**, e.g., in the case the materials of the first and third layers **404**, **407** have different etch rates.

(35) In some embodiments, reactive ions of a second angled etch **440** may be delivered into the first sidewall **416** of the opening **408** at a second non-zero angle 'β' relative to the perpendicular **420**, as further shown in FIG. **3**B. In this case, the second non-zero angle 'β' is greater than the first non-zero angle '⊖', which causes the second angled etch **440** to impact just the third layer **407** without impacting the first layer **404**. As a result, the lower portion **424** of the third layer **407** is further removed/recessed laterally (e.g., along the x-direction), as shown in FIG. **3**C. After the lower portion **424** of the third layer **407** is removed, the first plane defined by the sidewall **416** along the first layer **404** and the second plane defined by the sidewall **416** along the third layer **407** are oriented at different non-zero angles relative to the perpendicular **420**.

(36) In FIG. **3**D, a third reactive ion etch **450** may be performed along a second sidewall **417** of the opening **408**, similar to the first reactive ion etch **414**. That is, ions of the third reactive ion etch **450** may be delivered into the second sidewall **417** at a third non-zero angle 'α' relative to the perpendicular **420**. The third reactive ion etch **450** removes a portion of material from a lower section of the first and third layers **404**, **407** without significantly removing material from an upper section of the first and third layers **404**, **407**, as shown in FIG. **3**E. In some embodiments, the third non-zero angle of the third reactive ion etch **450** may be the same or similar to the first non-zero angle of first reactive ion etch **414**.

(37) Although not shown, reactive ions of a fourth angled etch may then be delivered into the second sidewall **417** of the opening **408** at a fourth non-zero angle relative to the perpendicular **420**. In some embodiments, the second non-zero angle is greater than the third non-zero angle 'α', and impacts just the third layer **407** without impacting the first layer **404**. As a result, the lower portion **424** of the third layer **407** is further removed/recessed laterally (e.g., along the x-direction). After the lower portion **424** of the third layer **407** is removed, a first plane defined by the sidewall **417** along the first layer **404** and a second plane defined by the sidewall **417** along the third layer **407** may be oriented at different non-zero angles relative to the perpendicular **420**. In some embodiments, the fourth non-zero angle of the fourth reactive ion etch may be the same or similar to the second non-zero angle of the second reactive ion etch **450**.
(38) FIGS. **4**A-**4**B depict a portion of a semiconductor device (hereinafter "device") **500** including a plurality of alternating vertical structures **505** and trenches **508**. Each of the vertical structures **505** may include a plurality of layers atop a substrate **702**. As shown, the plurality of layers may include a first layer **504** formed atop the substrate **702**, and a second layer **506** formed atop the first layer **504**. The first layer **504** may be a first material type, while the second layer **506** may be a second material type, different than the first material type.
(39) As further shown, reactive ions of a first angled etch **514** may be delivered into a first sidewall **516** of each vertical structure **505** at a first non-zero angle 'Θ' relative to a perpendicular **520** extending from a top surface **522** of the second layer **506**. In some embodiments, the first angled etch **514** may continue until the first sidewalls **516** are substantially vertical, as shown in FIG. **4**C. In other embodiments, the first angled etch **514** may continue until a notch or indentation **552** is formed in the first layer **504**, as shown in FIG. **4**D. In yet other embodiments, the indentation **552** may be formed as a result of a second angled etch process delivered at a different non-zero angle.
(40) FIGS. **5**A-**5**C demonstrate an approach for forming trenches without the use of an etch stop layer. In FIG. **5**A, reactive ions of a first angled ion etch **614** may be delivered to a device **600** having a plurality of alternating vertical structures **605** and trench material **611**. The vertical structures **605** and the trench material **611** may have different etch rates.
(41) As shown in FIG. **5**B, a portion of the trench material **611** is removed along an upper portion **628** of a first sidewall **616** of each vertical structure **605** as a result of the first angled ion etch **614**. Following a second angled ion etch **640**, a second portion of the trench material **611** is removed along the upper portion of a second sidewall **617** of each vertical structure **605**, as shown in FIG. **5**C. In this embodiment, a top surface **658** of the trench material **611** is recessed only partially relative to a top surface **622** of the second layer **606**, and without the use of an etch stop layer within the device **600**.
(42) Turning now to FIG. **6**A, an exemplary processing apparatus **700** for providing an ion exposure is illustrated. The processing apparatus **700** may be a compact plasma processing system operable to generate an ion beam shown as ions **710**. Ions **710** may be the same or similar to the ions of the angled etch processes described herein. More specifically, a substrate **702** may be exposed to reactive neutral species **724**, where the reactive neutral species **724** are derived from precursor gas composition used to generate the RIE plasma. The reactive neutral species **724** may arrive isotropically to the substrate **702**, where every portion of the different exposed surfaces of the substrate **702** are impacted by the reactive neutral species **724**. Notably, the present embodiments harness the principles of RIE processing where etching of a given surface is enhanced in the presence of ions. Notably, in accordance with the present embodiments, etching may take place just in regions of the substrate **702** impacted by the directional ions, i.e., in regions impacted by the ions **710**, while leaving other surfaces unetched.
(43) The ion beam may be extracted from a plasma **704** generated in a plasma chamber **703** by any known technique. The processing apparatus **700** may include an extraction plate **706** having an extraction aperture **708**, where the ions are extracted as an ion beam from the plasma **704** and directed to the substrate **702**. The substrate **702** may be the same as the substrates described herein.

As shown in FIG. **6**B, the extraction aperture **708** may be elongated along the Y-axis, providing a ribbon ion beam extending, for example, over an entire substrate along the direction parallel to the Y-axis. In various embodiments, the substrate **702** may be disposed on a substrate holder **715** and scanned along the X-axis to provide coverage at different regions of the substrate **702** or over the entirety of the substrate. In other embodiments, the extraction aperture **708** may have a different shape such as a square or circular shape.

(44) In some embodiments, the plasma chamber **703** may also serve as a deposition process chamber to provide material for depositing on the substrate **702** in the deposition operation preceding etching. The substrate holder **715** may further include a heater assembly **711** for selectively heating the substrate **702** to different temperatures in different regions within the X-Y plane for selectively changing the amount of depositing material as discussed above.

(45) During an ion exposure, reactive species may be provided or created in the plasma chamber **703** and may also impinge upon the substrate **702**. While various non-ionized reactive species may impinge upon all surfaces of substrate **702**, including different surfaces in one or more cavities, etching may take place in areas impacted by the ions **710**, as in known RIE processes, while little or no etching takes place in regions not impacted by ions **710**. Thus, referring to FIG. **3**B, for example, reactive ions of the second angled etch **440** may be delivered into the first sidewall **416** of the opening **408** at the second non-zero angle ‘β’ relative to the perpendicular **420**, which causes the second angled etch **440** to impact just the third layer **407** without impacting the first layer **404**. As a result, the lower portion **424** of the third layer **407** is further removed/recessed laterally.

(46) In further embodiments, directional etching of ions may be performed by rotating a substrate within the X-Y plane to any desired angle. Thus, a trench feature may be oriented with its long axis at a 45-degree angle with respect to the Y-axis while a ribbon beam directed to the trench feature has its axis oriented along the Y-axis as in FIG. **6**B.

(47) In additional embodiments, by scanning the substrate **702** with respect to the ion beam **710**, such as along the X-axis as generally shown in FIG. **6**B, the possibility is afforded to vary a directed etch across the substrate **702** to achieve location-specific directional selectivity of etching, so features within a certain region, such as region **712** of the substrate may be altered to one extent while features in another region, such as region **714**, are not altered or are altered to a different extent or in a different fashion. For example, the ion beam **710** may be present when region **712** is under extraction aperture **708**, while the ion beam is extinguished when the region **714** is under extraction aperture **708**.

(48) It is to be understood that the various layers, structures, and regions shown in the accompanying drawings are schematic illustrations. For ease of explanation, one or more layers, structures, and regions of a type commonly used to form semiconductor devices or structures may not be explicitly shown in a given drawing. This does not imply that any layers, structures, and/or regions not explicitly shown are omitted from the actual semiconductor structures.

(49) For the sake of convenience and clarity, terms such as “top,” “bottom,” “upper,” “lower,” “vertical,” “horizontal,” “lateral,” and “longitudinal” will be understood as describing the relative placement and orientation of components and their constituent parts as appearing in the figures. The terminology will include the words specifically mentioned, derivatives thereof, and words of similar import.

(50) While certain embodiments of the disclosure have been described herein, the disclosure is not limited thereto, as the disclosure is as broad in scope as the art will allow and the specification may be read likewise. Therefore, the above description is not to be construed as limiting. Instead, the above description is merely as exemplifications of particular embodiments. Those skilled in the art will envision other modifications within the scope and spirit of the claims appended hereto.

Claims

1. A method, comprising: providing a stack of layers of a semiconductor device; forming an opening through the stack of layers, the opening defined by a first sidewall and a second sidewall; and delivering ions into the first sidewall in a reactive ion etching process, wherein the ions are delivered at a first non-zero angle relative to a perpendicular extending from the stack of layers, wherein the reactive ion etching process removes a first portion of the stack of layers from just a lower section of the first sidewall, and wherein a plane defined by the lower section of the first sidewall is oriented at the first non-zero angle after the lower portion is removed.
2. The method of claim 1, wherein the reactive ion etching process does not remove a second portion of the stack of layers from an upper section of the first sidewall.
3. The method of claim 2, further comprising delivering ions into the second sidewall in a second reactive ion etching process, wherein the ions are delivered at a second non-zero angle relative to the perpendicular extending from the stack of layers, and wherein the second reactive ion etching process removes a first portion of the stack of layers from a lower section of the second sidewall without removing a second portion of the stack of layers from an upper section of the second sidewall.
4. The method of claim 3, wherein providing the stack of layers comprises: forming a first layer over a base layer; and forming a second layer over the first layer, wherein the first layer is a first material, and wherein the second layer is a second material different than the first material.
5. The method of claim 4, wherein providing the stack of layers further comprises: forming a third layer over the second layer; and forming a fourth layer over the third layer, wherein the third layer is a third material, and wherein the fourth layer is a fourth material different than the third material.
6. The method of claim 5, further comprising delivering ions into the first sidewall in a third reactive ion etching process, wherein the ions are delivered at a third non-zero angle relative to the perpendicular extending from the stack of layers, and wherein the third reactive ion etching process removes a lower portion of the third layer along the first sidewall without removing an upper portion of the third layer along the first sidewall.
7. The method of claim 6, further comprising delivering ions into the second sidewall in a fourth reactive ion etching process, wherein the ions are delivered at a fourth non-zero angle relative to the perpendicular extending from the stack of layers, and wherein the fourth reactive ion etching process removes a lower portion of the third layer along the second sidewall without removing an upper portion of the third layer along the second sidewall.
8. The method of claim 6, wherein the first non-zero angle is the same as the third non-zero angle.
9. A method for patterning opening sidewalls, the method comprising: providing a stack of layers of a semiconductor device, the stack of layers comprising: a first layer formed over a base layer; a second layer formed over the first layer; and a third layer formed over the second layer; forming an opening through the stack of layers; and directing ions into a sidewall of the first layer in a first reactive ion etch process, wherein the ions are delivered through the opening at a first non-zero angle relative to a perpendicular extending from the stack of layers, wherein the first reactive ion etch process removes a lower portion of the sidewall of the first layer, and wherein a plane defined by the sidewall of the first layer is oriented at the first non-zero angle after the lower portion of the sidewall of the first layer is removed.
10. The method of claim 9, further comprising directing ions into a sidewall of the second and third layers during the first reactive ion etch process, wherein the first layer is a first material, wherein the second layer is a second material-different than the first material, and wherein the third layer is a third material different than the second material.
11. The method of claim 10, wherein a first etch rate of the first layer is different than a second etch rate of the second material and different than a third etch rate of the third material, and wherein a difference between the first etch rate and the third etch rate causes a plane defined by the sidewall of the third layer to be oriented at a different angle than the plane defined by the sidewall of the

first layer.
12. The method of claim 10, wherein a first etch rate of the first layer is the same as a third etch rate of the third material, and wherein a plane defined by the sidewall of the third layer is oriented at a same angle as the plane defined by the sidewall of the first layer.
13. The method of claim 9, further comprising directing ions into a sidewall of the third layer in a second reactive ion etch process, wherein the ions are delivered through the opening at a second non-zero angle relative to the perpendicular extending from the stack of layers, wherein the second reactive ion etch process removes just a lower portion of the sidewall of the third layer.
14. The method of claim 13, wherein the second reactive ion etch causes a plane defined by the sidewall of the third layer to be oriented at the second non-zero angle after the lower portion of the sidewall of the third layer is removed, and wherein the first non-zero angle is different than the second non-zero angle.
15. The method of claim 13, wherein the second reactive ion etch process impacts the third layer without impacting the first layer.
16. A method, comprising: providing a stack of layers of a semiconductor device, the stack of layers comprising: a first layer formed over a base layer; a second layer formed over the first layer; and a third layer formed over the second layer; forming an opening through the stack of layers; and directing ions into a sidewall of the first layer in a first reactive ion etch process, wherein the ions are delivered through the opening at a first non-zero angle relative to a perpendicular extending from the substrate, wherein the first reactive ion etch process removes a lower portion of the sidewall of the first layer without removing an upper portion of the sidewall of the first layer, and wherein a plane defined by the sidewall of the first layer is oriented at the first non-zero angle after the lower portion of the sidewall of the first layer is removed.
17. The method of claim 16, further comprising directing ions into a sidewall of the second and third layers during the first reactive ion etch process, wherein the first layer is a first material, wherein the second layer is a second material different than the first material, and wherein the third layer is a third material-different than the second material.
18. The method of claim 17, wherein a first etch rate of the first layer is different than a second etch rate of the second material and different than a third etch rate of the third material, and wherein a difference between the first etch rate and the third etch rate causes a plane defined by the sidewall of the third layer to be oriented at a different angle than the plane defined by the sidewall of the first layer.
19. The method of claim 17, wherein a first etch rate of the first layer is the same as a third etch rate of the third material, and wherein a plane defined by the sidewall of the third layer is oriented at a same angle as the plane defined by the sidewall of the first layer.
20. The method of claim 16, further comprising directing ions into a sidewall of the third layer in a second reactive ion etch process, wherein the ions are delivered through the opening at a second non-zero angle relative to the perpendicular extending from the substrate, wherein the second reactive ion etch process removes just a lower portion of the sidewall of the third layer.